

Homogeneous Differential Equations

$$\boxed{1} \quad y' = \frac{x+y}{x}$$

$$y = ux : \quad u + xu' = 1 + u$$

$$xu' = 1 \quad \Rightarrow \quad u = \ln|x| + C$$

$$y = x(\ln|x| + C)$$

$$\boxed{2} \quad xy' = y + xe^{y/x}$$

$$u + xu' = u + e^u \quad \Rightarrow \quad xu' = e^u$$

$$-e^{-u} = \ln|x| + C$$

$$y = -x \ln(-\ln|x| - C)$$

$$\boxed{3} \quad y' = \frac{y-x}{y+x}$$

$$u + xu' = \frac{u-1}{u+1}$$

$$\frac{1}{2} \ln(1+u^2) + \arctan u = -\ln|x| + C$$

$$\ln(x^2 + y^2) = -2 \arctan \frac{y}{x} + C$$

$$\boxed{4} \quad (x^2 + y^2)y' = 2xy$$

$$u + xu' = \frac{2u}{1+u^2}$$

$$\int \frac{1+u^2}{u(1-u^2)} du = \int \frac{dx}{x}$$

$$\left| \frac{u}{1-u^2} \right| = Cx \quad \Rightarrow \quad x^2 - y^2 = Cy$$

$$\boxed{5} \quad y' = \frac{y}{x} + \frac{x}{y}$$

$$u + xu' = u + \frac{1}{u} \quad \Rightarrow \quad xu' = \frac{1}{u}$$

$$u du = \frac{dx}{x} \quad \Rightarrow \quad \frac{u^2}{2} = \ln|x| + C$$

$$y^2 = x^2(2 \ln |x| + C)$$

$$\boxed{6} \quad xy' = y + x \cos^2 \frac{y}{x}$$

$$u + xu' = u + \cos^2 u \quad \Rightarrow \quad xu' = \cos^2 u$$

$$\sec^2 u \, du = \frac{dx}{x} \quad \Rightarrow \quad \tan u = \ln |x| + C$$

$$\tan \frac{y}{x} = \ln |x| + C$$

$$\boxed{7} \quad xy' = 2x - y$$

$$u + xu' = 2 - u \quad \Rightarrow \quad xu' = 2 - 2u$$

$$\frac{du}{1-u} = \frac{2 \, dx}{x} \quad \Rightarrow \quad -\ln |1-u| = 2 \ln |x| + C$$

$$y = x - \frac{C}{x}$$

$$\boxed{8} \quad (x + 2y) \, dx - x \, dy = 0$$

$$u + xu' = 1 + 2u \quad \Rightarrow \quad xu' = 1 + u$$

$$\frac{du}{1+u} = \frac{dx}{x} \quad \Rightarrow \quad \ln |1+u| = \ln |x| + C$$

$$y = Cx^2 - x$$

$$\boxed{9} \quad x^2 y' = y^2 + xy$$

$$u + xu' = u^2 + u \quad \Rightarrow \quad xu' = u^2$$

$$\frac{du}{u^2} = \frac{dx}{x} \quad \Rightarrow \quad -\frac{1}{u} = \ln |x| + C$$

$$y = \frac{-x}{\ln |x| + C}$$

$$\boxed{10} \quad y' = \frac{y}{x} \left(\ln \frac{y}{x} + 1 \right)$$

$$u + xu' = u \ln u + u \quad \Rightarrow \quad xu' = u \ln u$$

$$\frac{du}{u \ln u} = \frac{dx}{x} \quad \Rightarrow \quad \ln |\ln u| = \ln |x| + C$$

$$y = x e^{C x}$$

Homework

$$\boxed{1} \quad y' = \frac{2x+y}{x}$$

$$\boxed{2} \quad xy' = y + \sqrt{x^2 + y^2}$$

$$\boxed{3} \quad (x - y)y' = x + y$$

$$\boxed{4} \quad y' = \frac{y}{x} + \cos \frac{y}{x}$$

$$\boxed{5} \quad (x^2 - y^2) dx + 2xy dy = 0$$

$$\boxed{6} \quad xy' = y + x \sin \frac{y}{x}$$

$$\boxed{7} \quad y' = \frac{y^2}{x^2} + 2$$

$$\boxed{8} \quad (2x - 3y) dx + x dy = 0$$

$$\boxed{9} \quad x^2 y' = y^2 - xy$$

$$\boxed{10} \quad y' = \frac{y}{x} \ln \frac{y}{x}$$